

handsOn2 -task1

17: Filter Courses by Department

```
>>> Course.objects.filter(
...     department__name="Computer Science"
... )
<QuerySet [<Course: Data Structures>, <Course: Operating Systems>]>
>>>
```

18: Count Courses in Every Department

```
>>> for dept in Department.objects.annotate(
...     course_count=Count("course")):
...     print(dept.name, dept.course_count)
...
Computer Science 0
Information Technology 0
Computer Science 2
Information Technology 2
>>>
```

Using `.values()`

```
>>> Department.objects.annotate(
...     course_count=Count("course")
... ).values("name", "course_count")
<QuerySet [{ 'name': 'Computer Science', 'course_count': 0 }, { 'name': 'Information Technology', 'course_count': 0 },
{ 'name': 'Computer Science', 'course_count': 2 }, { 'name': 'Information Technology', 'course_count': 2 }]>
>>>
```

TASK 2

```
(venv) PS C:\Users\Admin\coursemanager> python manage.py createsuperuser
Username (leave blank to use 'admin'):
Email address: admin@django.com
Password:
Password (again):
```

```
Superuser created successfully.
(venv) PS C:\Users\Admin\coursemanager>
```

22: Register Models

admin.py

```
from django.contrib import admin
from .models import Department, Course, Student, Enrollment

admin.site.register(Department)
admin.site.register(Course)
admin.site.register(Student)
admin.site.register(Enrollment)
```

